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GENERAL CHEMISTRY I

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Office hour: by appointment

Textbook: Chemistry: The Central Science, By Theodore E. Brown

Expected Learning Outcomes

1. Understand the basic principles of general chemistry
2. Be able to explain basic, everyday chemical phenomena and apply your knowledge to solving real-world problems
3. Be able to explain why chemistry is essential and how it applies to everyday life

5 ways to fail this course

- Use communication devices in the exams
- Do not fully participate in the lab sections
- Take more than 3 absent days with or without the faculty's permission
- Fake lab data
- Plagiarize

What's your own definition
of chemistry?

GENERAL CHEMISTRY I



Chapter 1 Introduction: Matter and Measurement

Contents

- 1-1 The Study of Chemistry
- 1-2 Classification of Matter
- 1-3 Properties of Matter
- 1-4 Units of Measurement: SI (Metric) System
- 1-5 Uncertainties in Scientific Measurements
- 1-6 Significant Figures

Nutrition Facts

Serving Size 12 fl oz (360 mL)

Servings Per Container 6

Amount Per Serving

Calories 140

% Daily Value*

Total Fat 0g 0%

Sodium 45mg 2%

Total Carbohydrate 39g 13%

Sugars 39g

Protein 0g

Not a significant source of calories from fat, saturated fat, trans fat, cholesterol, dietary fiber, vitamin A, vitamin C, calcium and iron.

*Percent Daily Values are based on a 2,000 calorie diet.

CAR
COL
FLAV
CAFF
©20
CON
CALL
WWW

Coke®



Chemistry is all around us

Energy

Solar panels are composed of specially treated silicon.



Biochemistry

The flash of the firefly results from a chemical reaction in the insect.



Technology

LED's (light emitting diodes) are formed from elements such as gallium, arsenic and phosphorus.



Medicine

Connectors and tubing for medical procedures such as intravenous injections are made from plastics highly resistant to chemical attack.



Chemistry

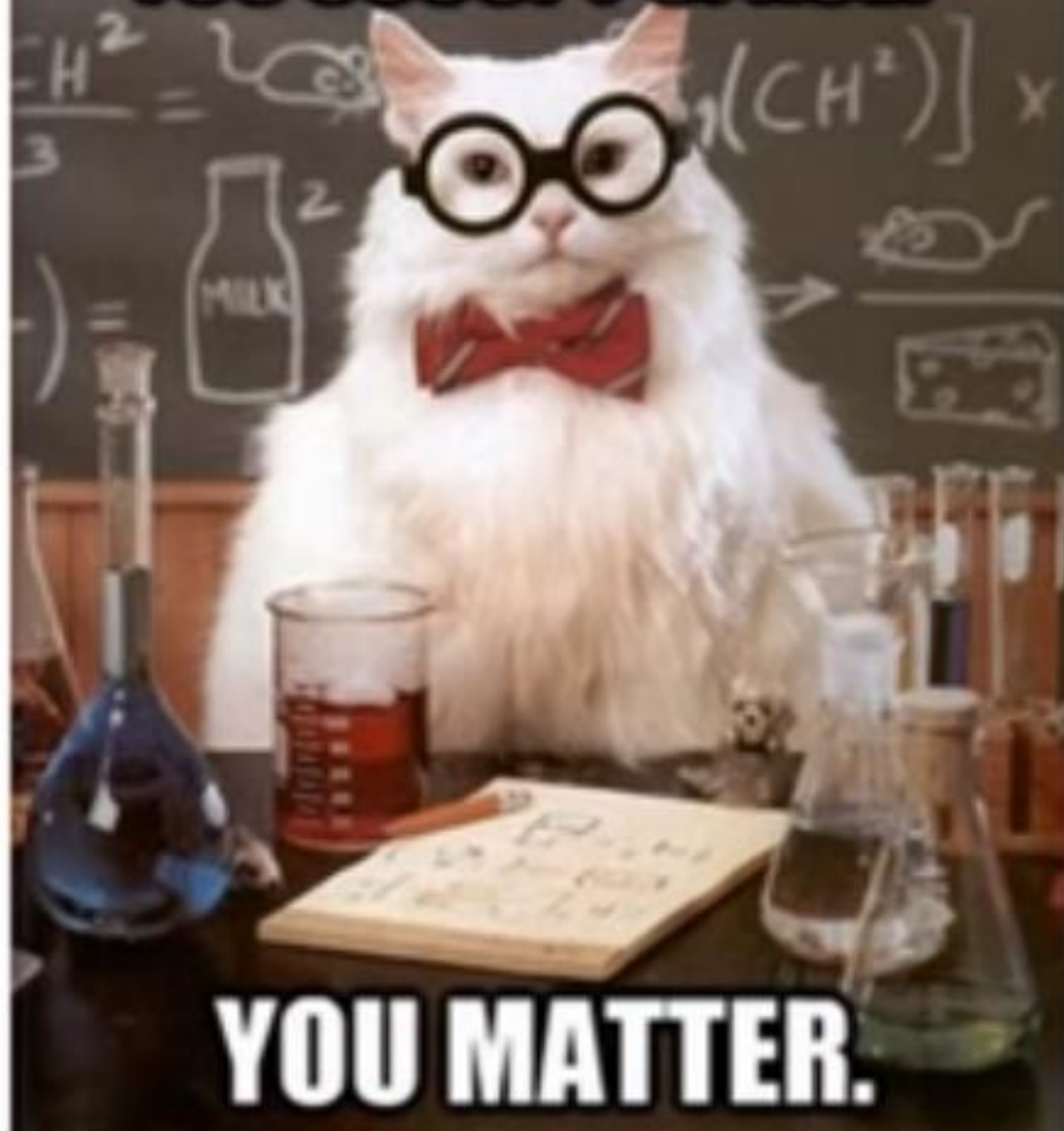
Chemistry is central to our understanding of the world around us

1-1 The Study of Chemistry



Chemistry is the study of matter
and the changes that matter undergoes

**DO YOU HAVE MASS? DO
YOU OCCUPY SPACE?**



YOU MATTER.

The Periodic Table of the Elements

The Periodic Table of the Elements

group 1

period 1

1

1.00794

H

Hydrogen

2

6.941

Li

Lithium

3

9.012182

Be

Beryllium

4

22.98976

Na

Sodium

11

24.3050

Mg

Magnesium

12

39.0983

K

Potassium

19

40.078

Ca

Calcium

20

44.95591

Sc

Scandium

21

47.867

Ti

Titanium

22

50.9415

V

Vanadium

23

51.9962

Cr

Chromium

24

54.93804

Mn

Manganese

25

55.845

Fe

Iron

26

58.93319

Co

Cobalt

27

58.6934

Ni

Nickel

28

63.546

Cu

Copper

29

65.38

Zn

Zinc

30

69.723

Ga

Gallium

31

72.64

Ge

Germanium

32

74.9216

As

Arsenic

33

78.96

Se

Selenium

34

79.904

Br

Bromine

35

83.798

Kr

Krypton

36

85.4678

Rb

Rubidium

37

87.62

Sr

Strontium

38

88.90585

Y

Yttrium

39

91.224

Zr

Zirconium

40

92.90638

Nb

Niobium

41

95.96

Mo

Molybdenum

42

(98)

Tc

Technetium

43

101.07

Ru

Ruthenium

44

102.9055

Rh

Rhodium

45

106.42

Pd

Palladium

46

107.8682

Ag

Silver

47

112.414

Cd

Cadmium

48

114.818

In

Indium

49

118.710

Sn

Tin

50

121.760

Sb

Antimony

51

127.60

Te

Tellurium

52

126.9044

I

Iodine

53

131.293

Xe

Xenon

54

132.9054

Cs

Cesium

55

137.327

Ba

Barium

56

174.9668

Lu

Lutetium

71

178.49

Hf

Hafnium

72

180.9478

Ta

Tantalum

73

183.84

W

Tungsten

74

186.207

Re

Rhenium

75

190.23

Os

Osmium

76

192.217

Ir

Iridium

77

195.084

Pt

Platinum

78

196.9665

Au

Gold

79

200.59

Hg

Mercury

80

204.3833

Tl

Thallium

81

207.2

Pb

Lead

82

208.9804

Bi

Bismuth

83

(210)

Po

Polonium

84

(210)

At

Astatine

85

(220)

Rn

Radon

86

(223)

Fr

Francium

87

(226)

Ra

Radium

88

(262)

Lr

Lanthanum

103

(261)

Rf

Rutherfordium

104

(262)

Db

Dubnium

105

(266)

Sg

Seaborgium

106

(264)

Bh

Bohrium

107

(277)

Hs

Hassium

108

(268)

Mt

Mtnerium

109

(271)

Ds

Darmstadtium

110

(272)

Rg

Roentgenium

111

(285)

Cn

Copernicium

112

(284)

Uut

Ununium

113

(289)

Fl

Flerovium

114

(288)

Uup

Ununpentium

115

(292)

Lv

Livermorium

116

Uus

Ununseptium

117

(294)

Uuo

Ununoctium

118

4.002602

He

Helium

alkali metals

alkaline metals

other metals

transition metals

lanthanoids

actinoids

metalloids

nonmetals

halogens

noble gases

unknown elements

radioactive elements have masses in parentheses

atomic mass or most stable mass number

1st ionization energy in kJ/mol

chemical symbol

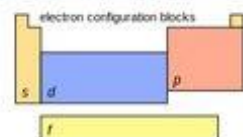
name

electron configuration

atomic number

electronegativity

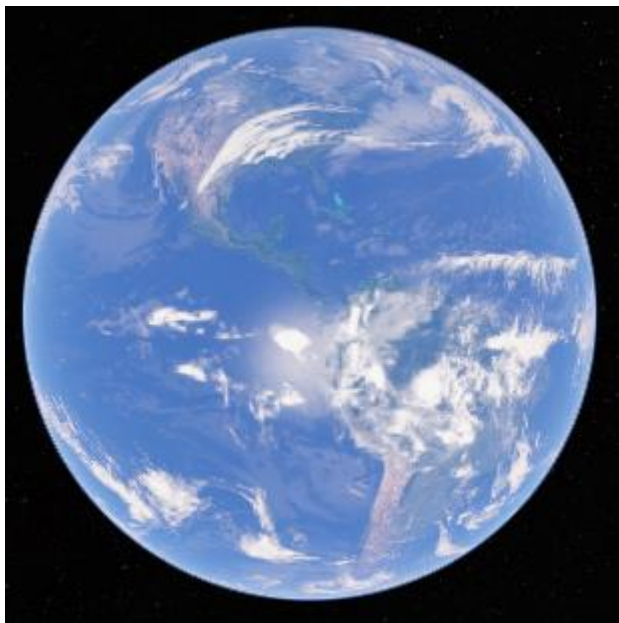
oxidation states most common are bold



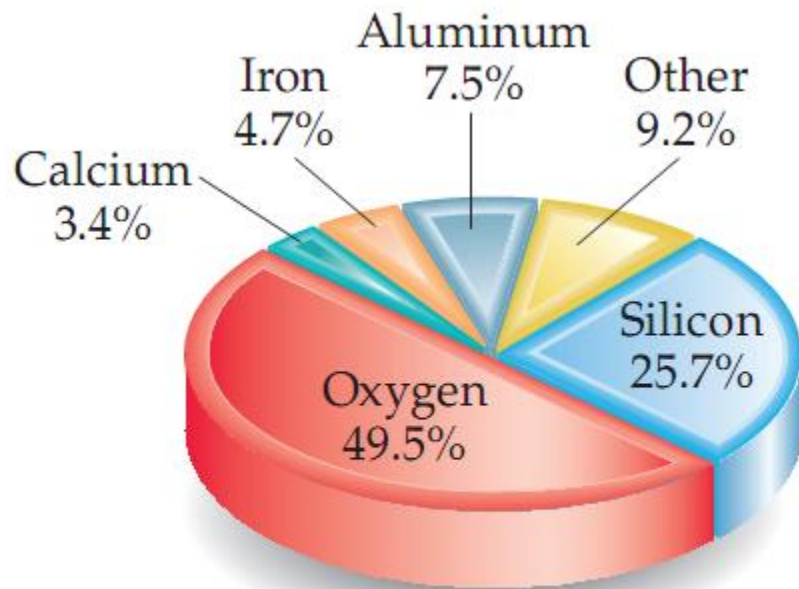
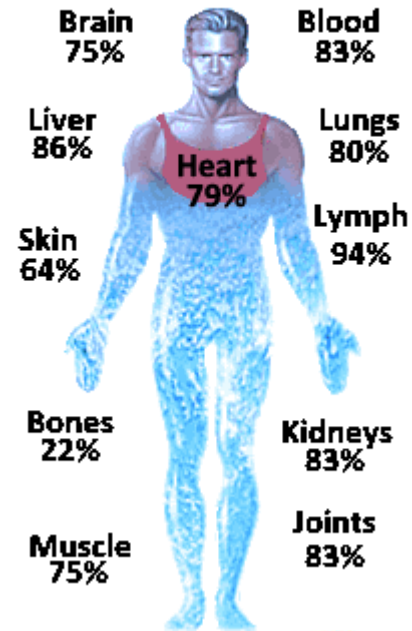
notes

- as of yet, elements 113, 115, 117 and 118 have no official name designated by the IUPAC.
- 1 kJ/mol = 96.485 eV.
- all elements are implied to have an oxidation state of zero.

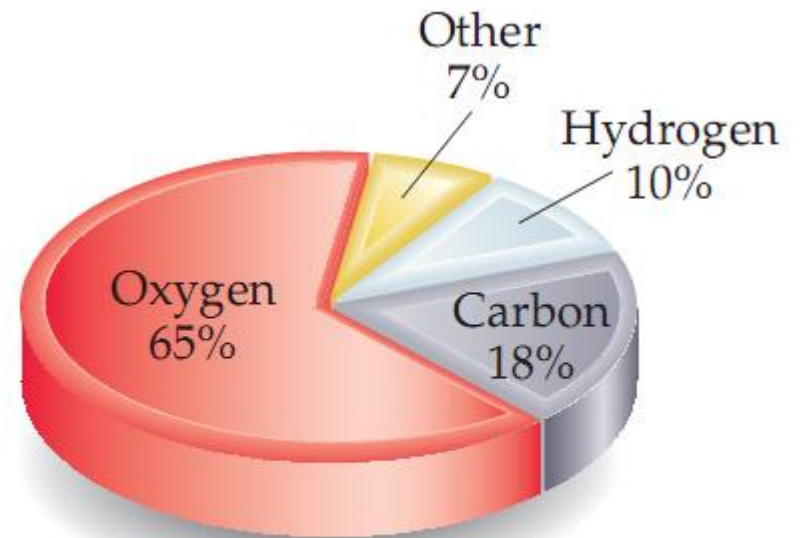
138.9054 57	140.116 58	140.9076 59	144.242 60	(145) 61	150.36 62	151.964 63	157.25 64	158.9253 65	162.500 66	164.9303 67	167.259 68	168.9342 69	173.054 70
La Lanthanum (94) 57	Ce Cerium (94) 58	Pr Praseodymium (94) 59	Nd Neodymium (94) 60	Pm Promethium (94) 61	Sm Samarium (94) 62	Eu Europium (94) 63	Gd Gadolinium (94) 64	Tb Terbium (94) 65	Dy Dysprosium (94) 66	Ho Holmium (94) 67	Er Erbium (94) 68	Tm Thulium (94) 69	Yb Ytterbium (94) 70
(227) 89	232.0380 90	231.0358 91	238.0289 92	(237) 93	(244) 94	(243) 95	(247) 96	(247) 97	(251) 98	(252) 99	(257) 100	(258) 101	(259) 102
Ac Actinium (94) 89	Th Thorium (94) 90	Pa Protactinium (94) 91	U Uranium (94) 92	Np Neptunium (94) 93	Pu Plutonium (94) 94	Am Americium (94) 95	Cm Curium (94) 96	Bk Berkelium (94) 97	Cf Californium (94) 98	Es Einsteinium (94) 99	Fm Fermium (94) 100	Md Mendelevium (94) 101	No Nobelium (94) 102



Our Body is 60% ~70% Water



Earth's crust



Human body

Atoms

Atoms: the almost infinitesimally small building blocks of matter

The Scale of the Universe 2

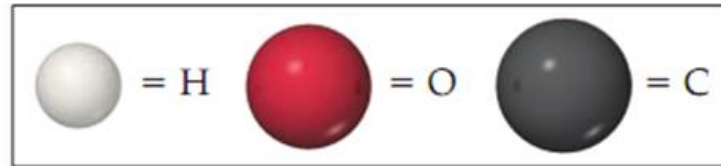
Molecules and compounds

Molecule: two or more atoms are joined in specific shapes

Homonuclear molecule



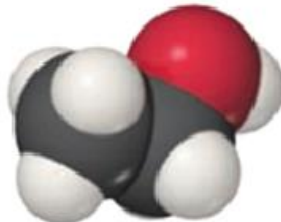
Oxygen



Compound



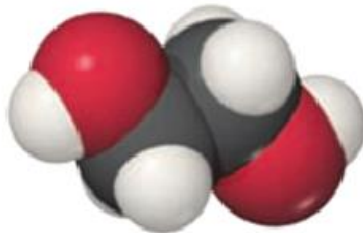
Water



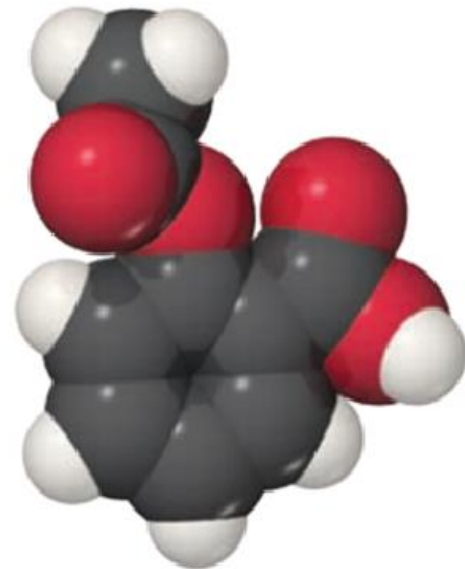
Ethanol



Carbon dioxide

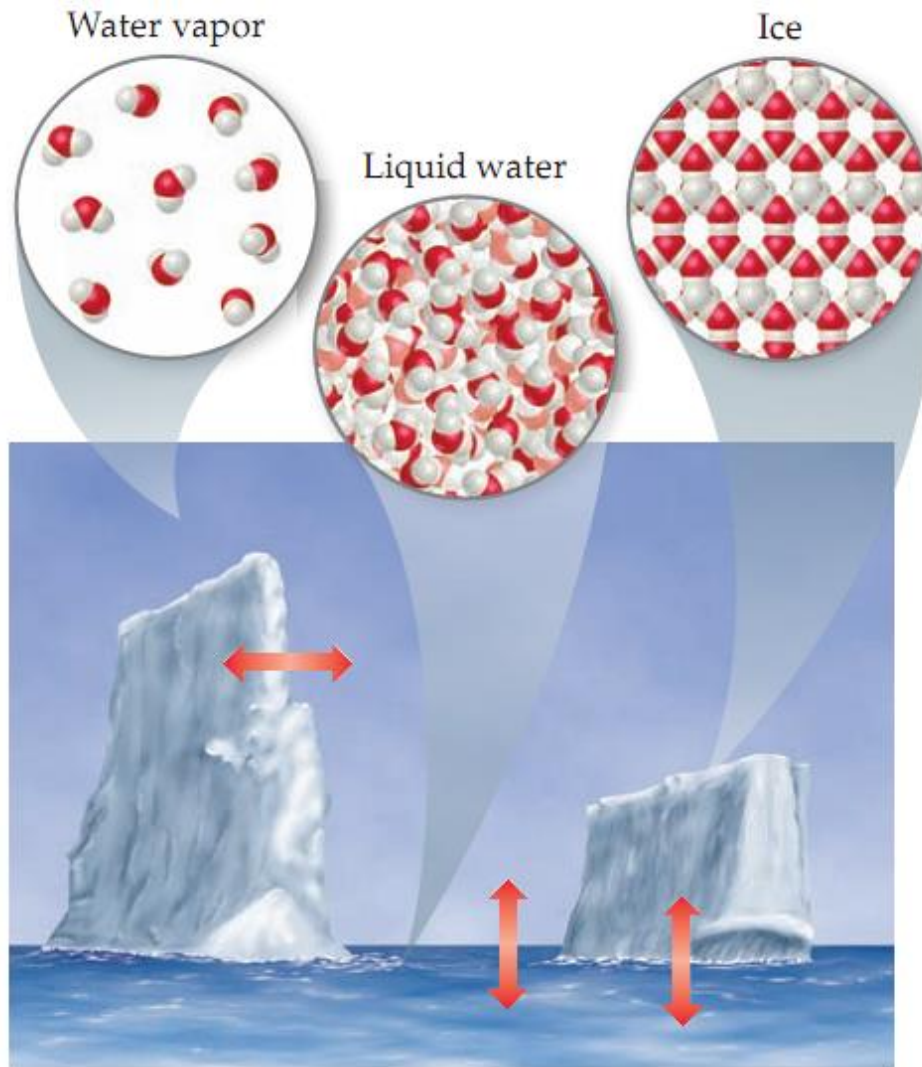


Ethylene glycol



Aspirin

1-2 Classification of Matter

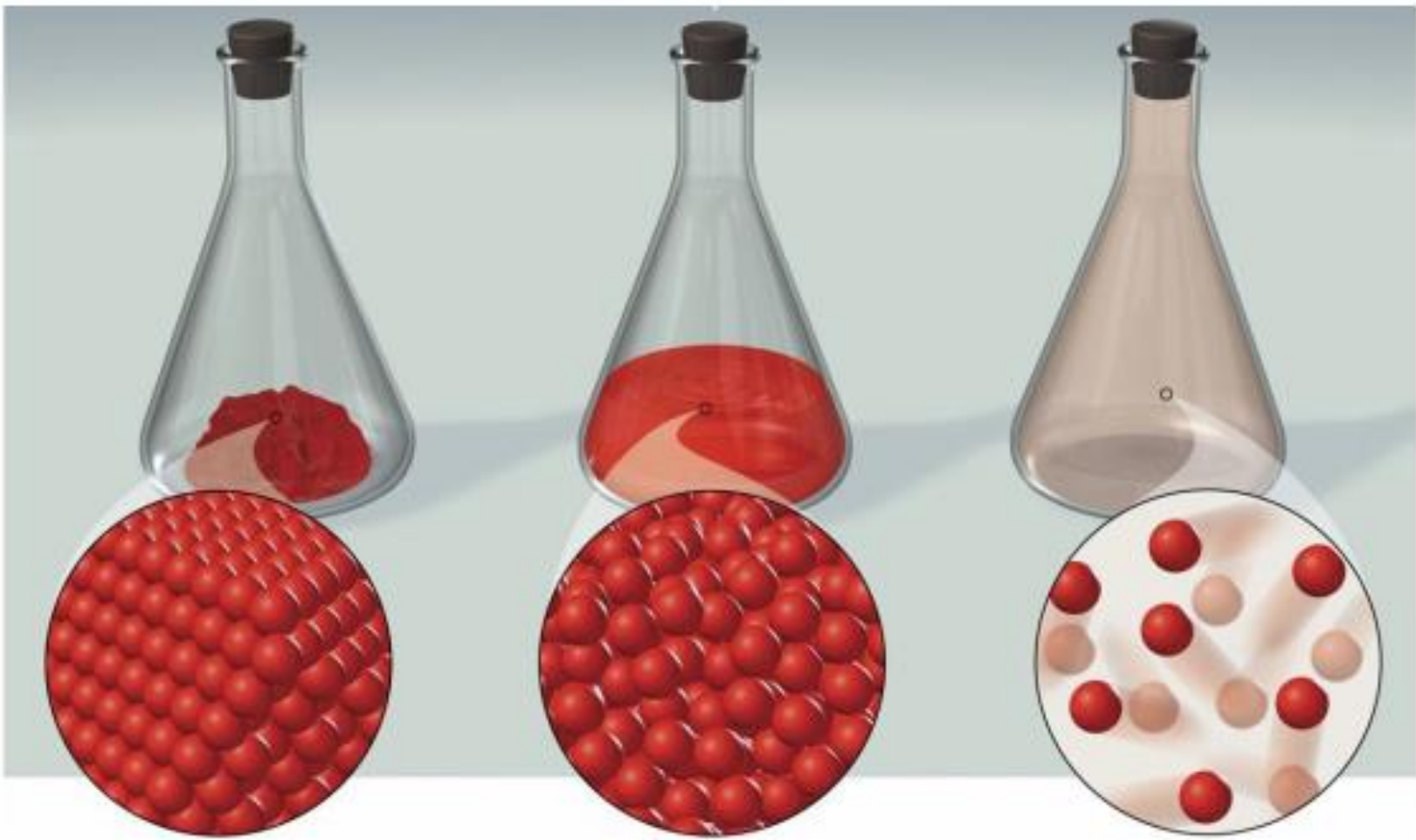


In which state of matter (gas, liquid or solid) are individual molecular furthest apart?

In which one they are the second furthest apart?

In which state are they closest together?

Physical States of Matter



Solids

have a fixed shape
and volume

not compressible

Liquids

conform to the shape
of a container, but not
the volume

not compressible

Gases

take up the shape
and volume of a
container

compressible

Pure substances

Pure substances: matter that has distinct properties and a composition that does not vary from sample to sample



Diamond (carbon, C)

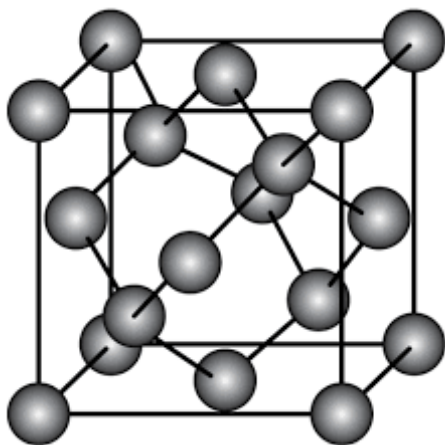
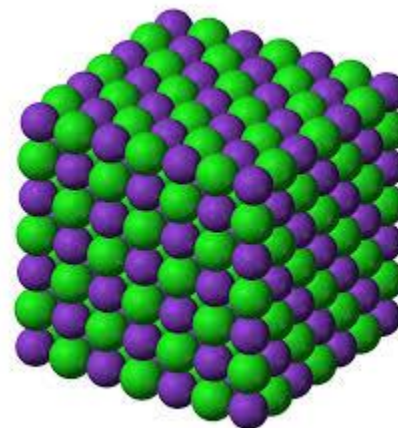


Table salt (sodium chloride, NaCl)



Mixtures

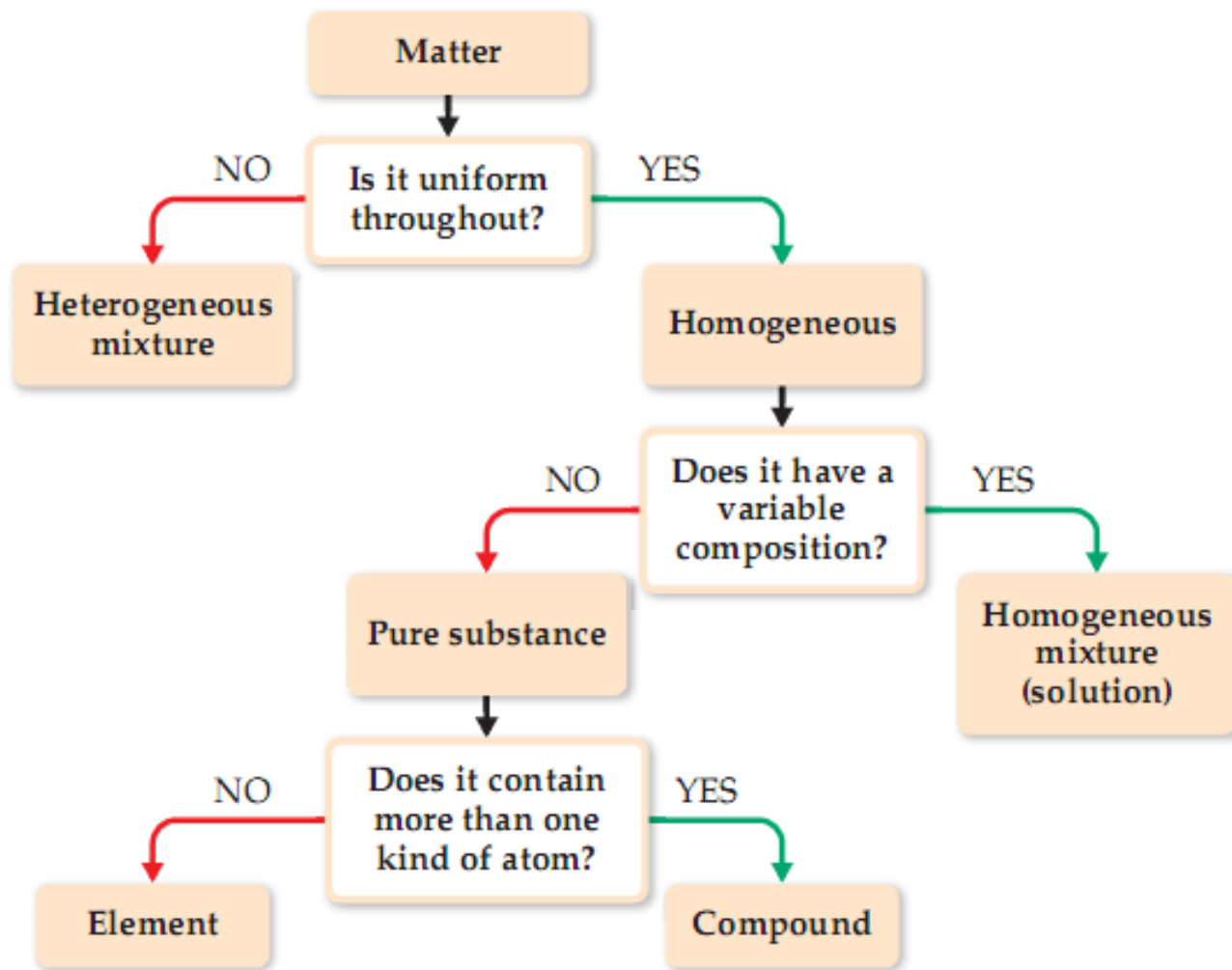
Mixture: collection of two or more pure substances



rocks
a heterogeneous mixtures

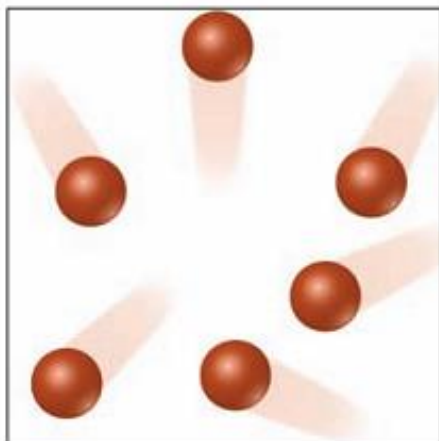


copper(II) sulfate aqueous solution
a homogeneous mixtures

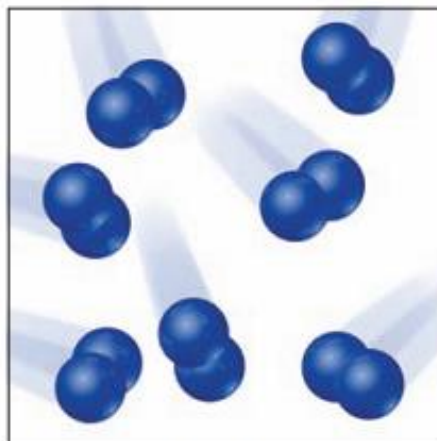


- **Matter** is made of atoms.
- **Atoms** are differentiated as **elements** (118 elements known).
- **Compounds** are comprised of two or more elements.
- **Molecules** are the smallest units of compounds.

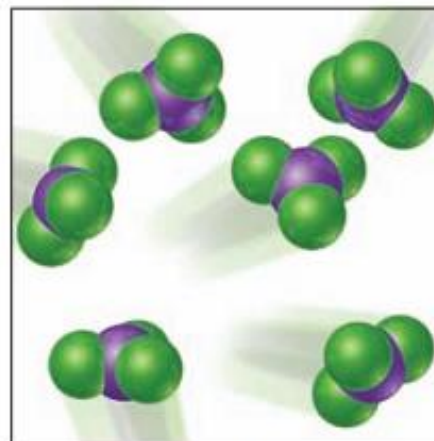
How do the molecules of a compound differ from the molecules of an element?



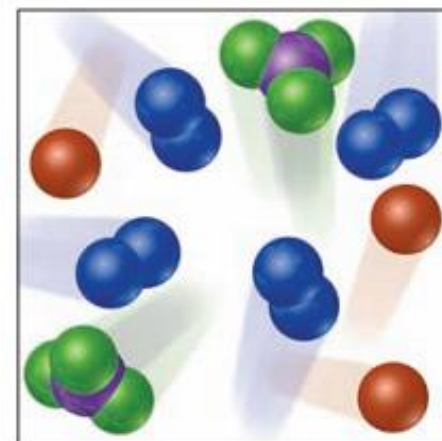
(a) Atoms of an element



(b) Molecules of an element



(c) Molecules of a compound

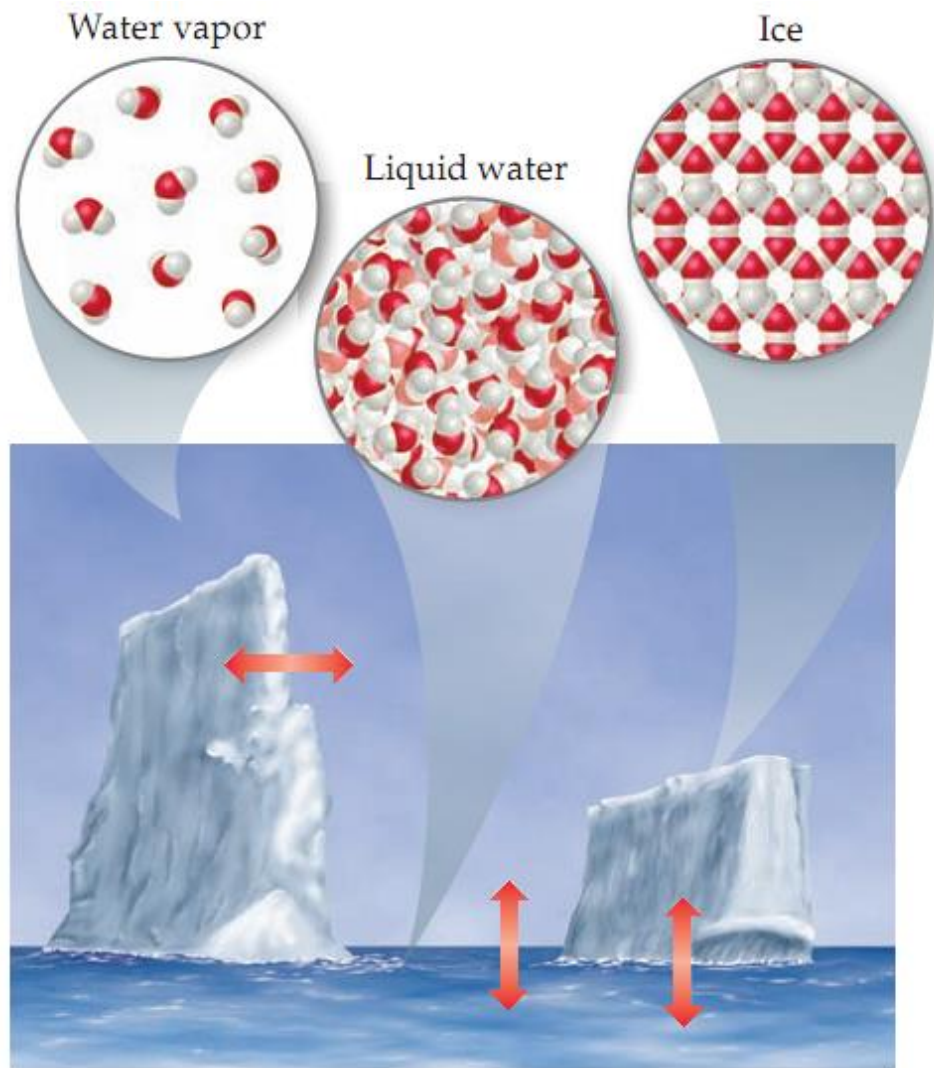


(d) Mixture of elements and a compound

Only one kind of atom is in any element.

Compounds must have at least two kinds of atoms.

1-3 Properties of Matter

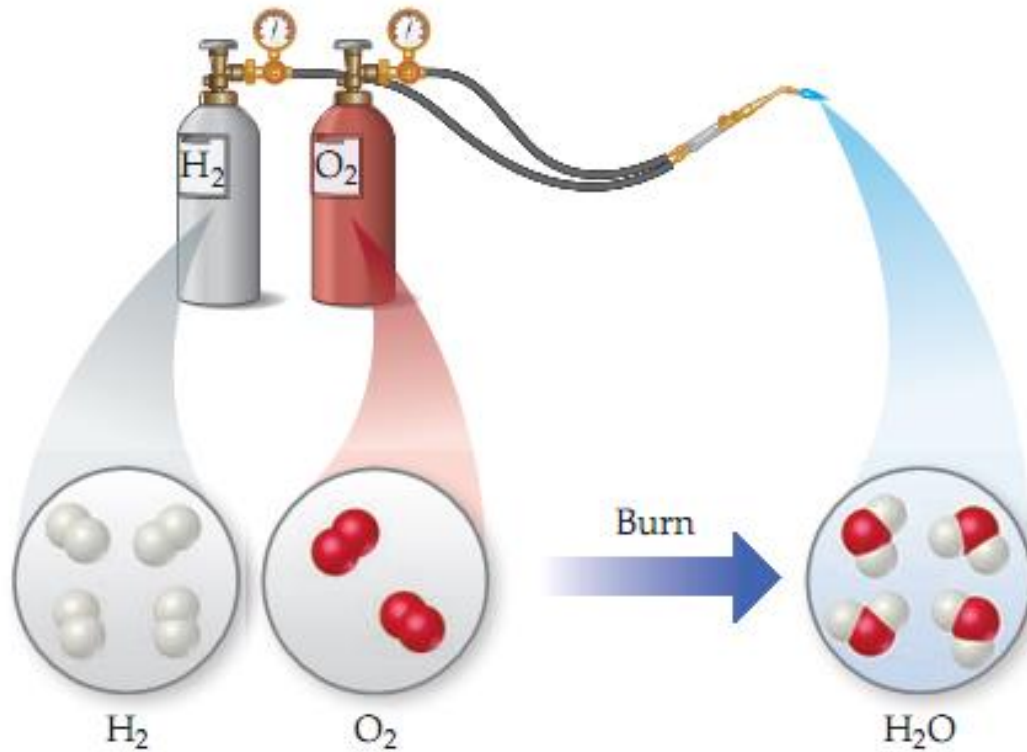


Physical
change

Physical appearance changes but chemical composition does not

Chemical change

Chemical change (also called a **chemical reaction**): a substance is transformed into a chemically different substance



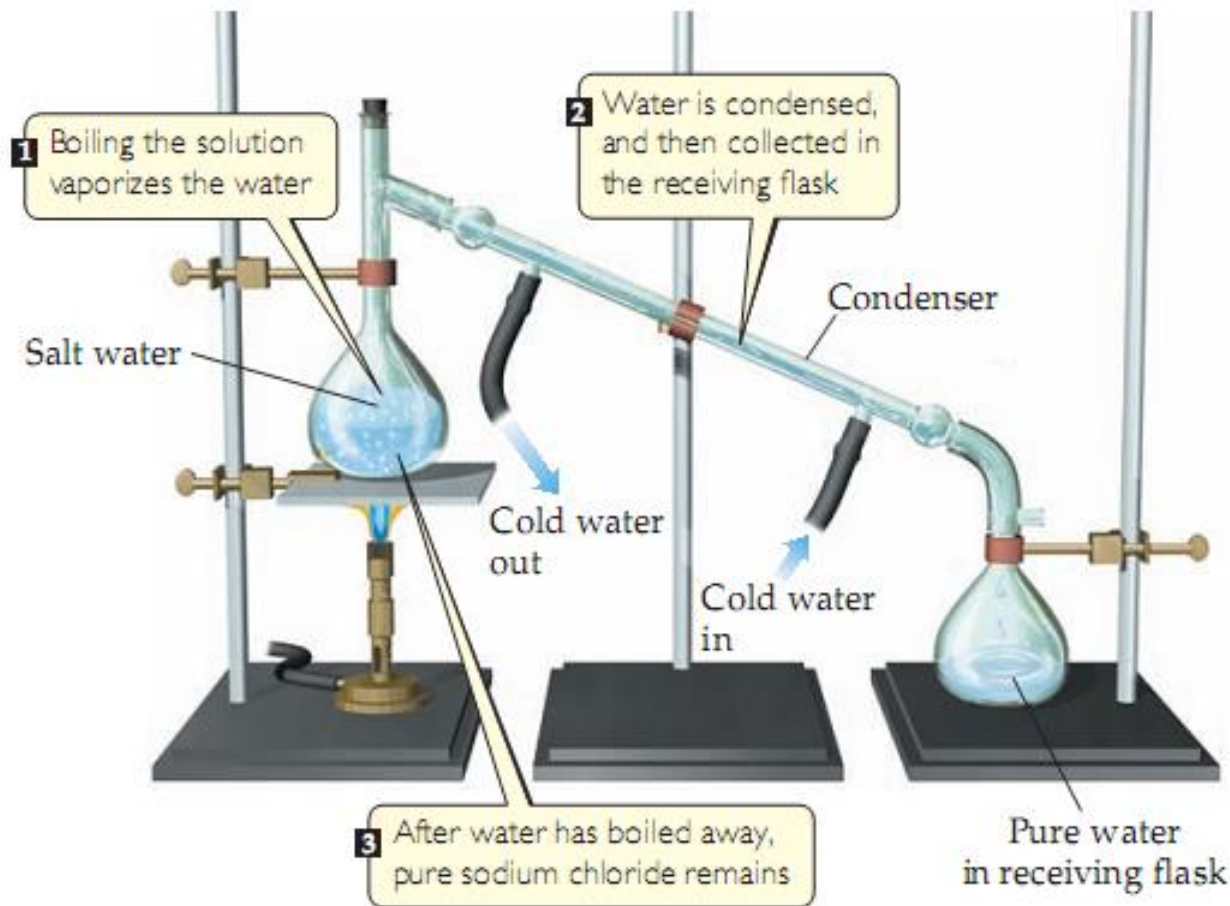
Chemical change – formation of a new compound (water)



Chemical change – decomposition of a compound (ammonium chromate)

Separation of Mixtures

Physical change : a substance changes its physical appearance but not its composition



Separation of a mixture (salt water) by distillation

EXAMPLE

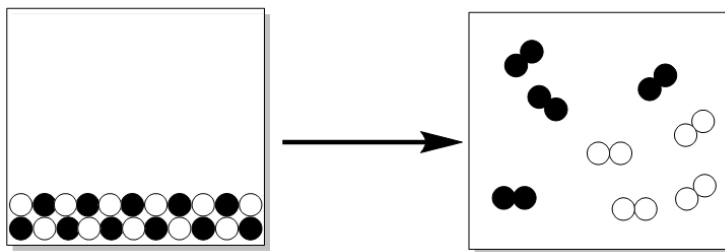
Which of the following represents a physical change?

- a. Alkanes burn spontaneously.*
- b. The sublimation of CO_2 .*
- c. $2 \text{H}_2 (\text{g}) + \text{O}_2 (\text{g}) \rightarrow 2 \text{H}_2\text{O} (\text{g})$*
- d. The rusting of a car.*

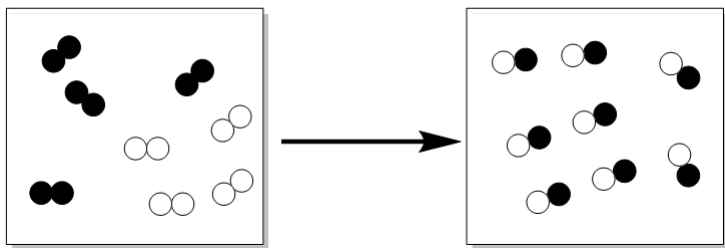
EXAMPLE

Determine which of the following represents a physical change and a chemical change.

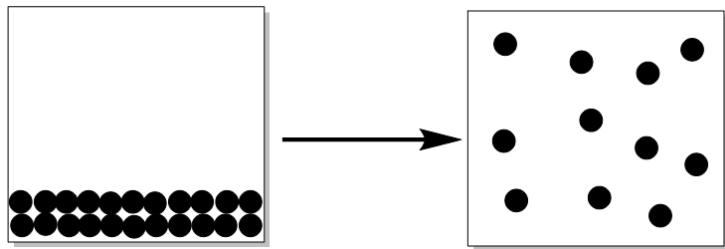
a.



b.

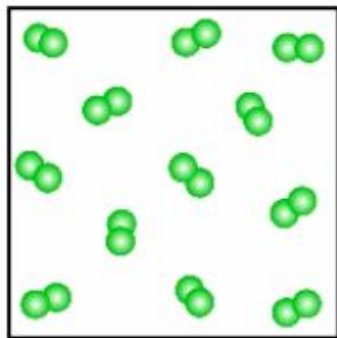


c.

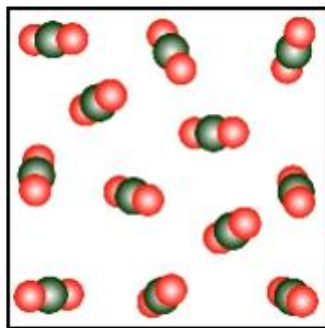


EXAMPLE

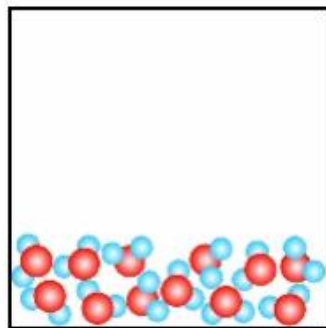
Answer each of the following questions based on the images provided below.



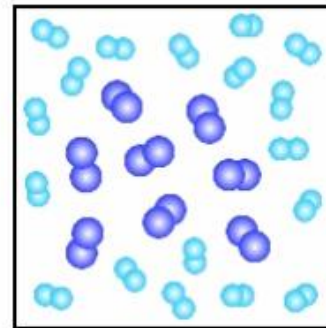
1



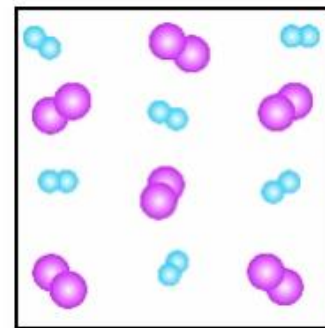
2



3



4



5

- a) Which images represents an elemental gas?
- b) Which images represents a heterogeneous mixture?
- c) Which images represents a liquid?

EXAMPLE

Identify each of the following homogeneous or heterogeneous mixture

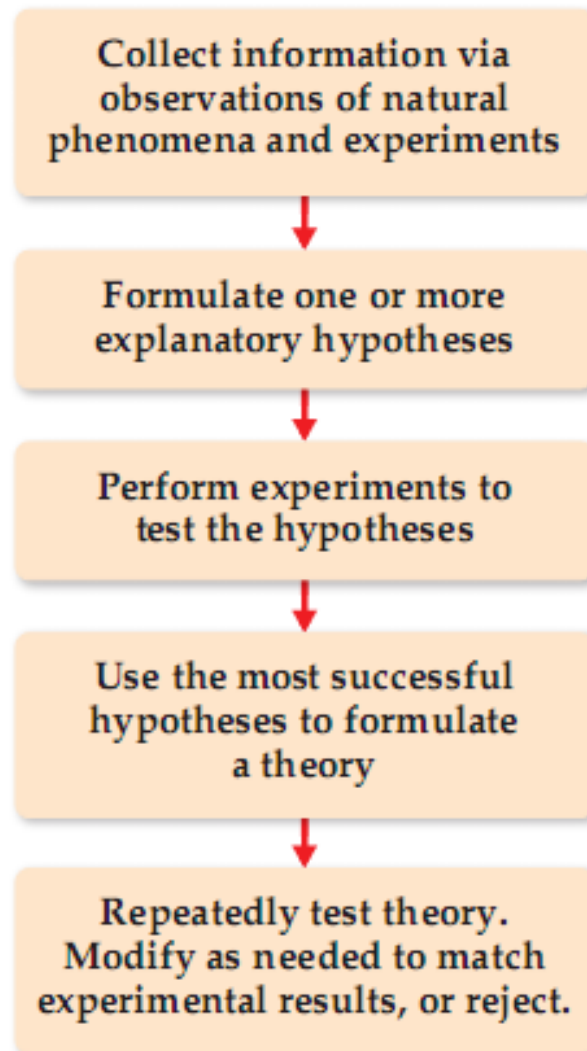
a) Wood

b) Tap water

c) Dirt

d) Air

1-4 The Measurement of Matter



Scientific Method

Table 1.4 SI Base Units

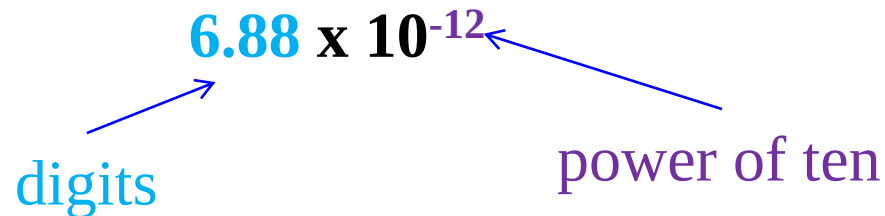
Physical Quantity	Name of Unit	Abbreviation
Mass	Kilogram	kg
Length	Meter	m
Time	Second	s or sec
Temperature	Kelvin	K
Amount of substance	Mole	mol
Electric current	Ampere	A or amp
Luminous intensity	Candela	cd

Table 1.5 Prefixes Used in the Metric System and with SI Units

Prefix	Abbreviation	Meaning	Example	
Peta	P	10^{15}	1 petawatt (PW)	$= 1 \times 10^{15} \text{ watts}^a$
Tera	T	10^{12}	1 terawatt (TW)	$= 1 \times 10^{12} \text{ watts}$
Giga	G	10^9	1 gigawatt (GW)	$= 1 \times 10^9 \text{ watts}$
Mega	M	10^6	1 megawatt (MW)	$= 1 \times 10^6 \text{ watts}$
Kilo	k	10^3	1 kilowatt (kW)	$= 1 \times 10^3 \text{ watts}$
Deci	d	10^{-1}	1 deciwatt (dW)	$= 1 \times 10^{-1} \text{ watt}$
Centi	c	10^{-2}	1 centiwatt (cW)	$= 1 \times 10^{-2} \text{ watt}$
Milli	m	10^{-3}	1 milliwatt (mW)	$= 1 \times 10^{-3} \text{ watt}$
Micro	μ^b	10^{-6}	1 microwatt (μW)	$= 1 \times 10^{-6} \text{ watt}$
Nano	n	10^{-9}	1 nanowatt (nW)	$= 1 \times 10^{-9} \text{ watt}$
Pico	p	10^{-12}	1 picowatt (pW)	$= 1 \times 10^{-12} \text{ watt}$
Femto	f	10^{-15}	1 femtowatt (fW)	$= 1 \times 10^{-15} \text{ watt}$
Atto	a	10^{-18}	1 attowatt (aW)	$= 1 \times 10^{-18} \text{ watt}$
Zepto	z	10^{-21}	1 zeptowatt (zW)	$= 1 \times 10^{-21} \text{ watt}$

Scientific Notation

Scientific notation is used to turn small or large, inconvenient numbers into manageable ones



The diagram shows the scientific notation 6.88×10^{-12} . A blue arrow points from the word "digits" to the number 6.88. A purple arrow points from the words "power of ten" to the 10^{-12} part of the expression.

$$6.88 \times 10^{-12}$$

digits

power of ten

the decimal point placed after the first digit

Six point eighty eight times ten to the minus twelfth

EXAMPLE

Convert the following numbers into scientific notation

a. 377,000

b. 0.000101

c. 707.82

d. 161.00×10^7

Mass

- Mass is the **quantity** of matter in an object.
- Weight is the force of gravity on an object

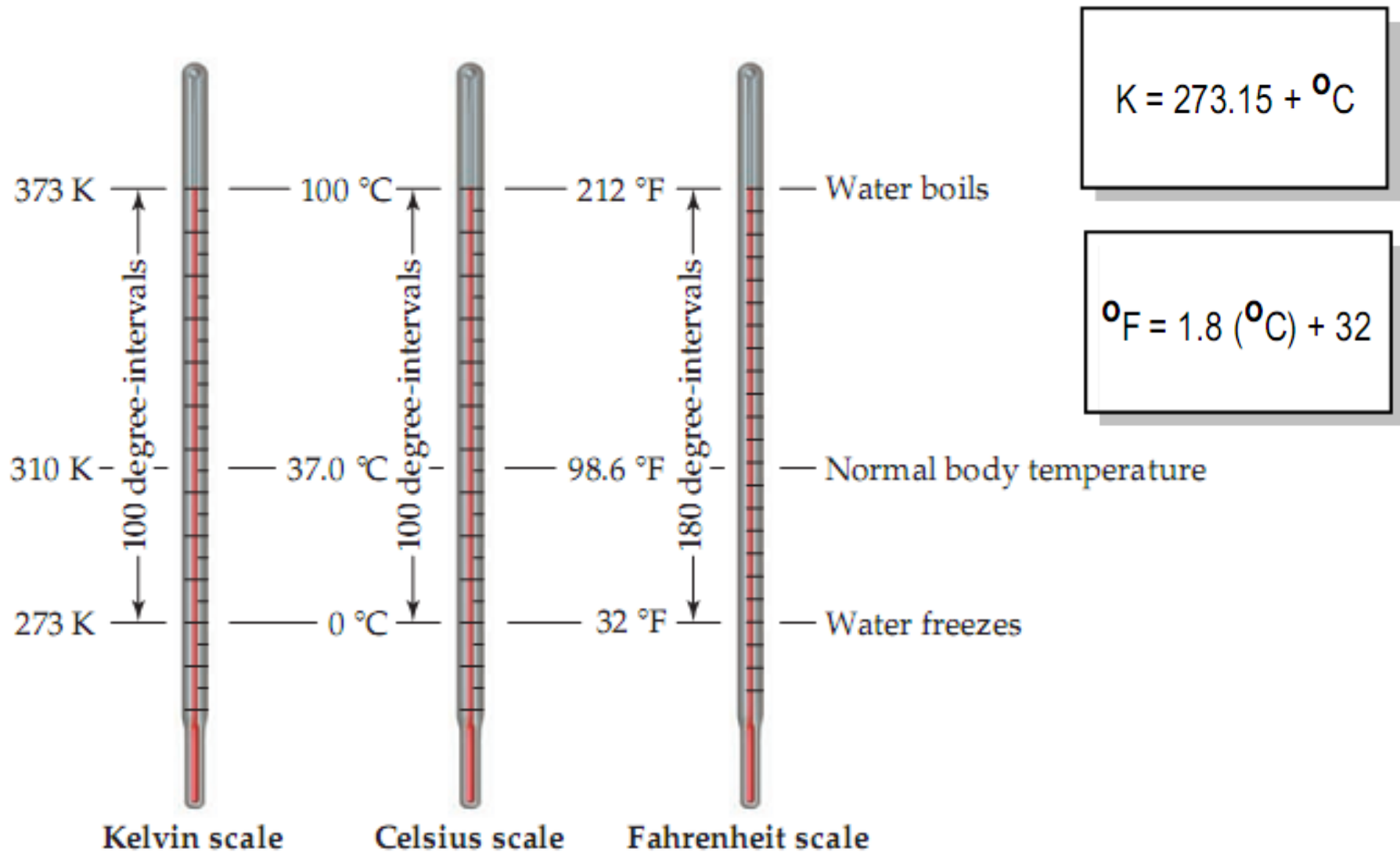
$$W \propto m$$

$$W = g \cdot m$$



Temperature

Temperature is a measure of the hotness or coldness of an object

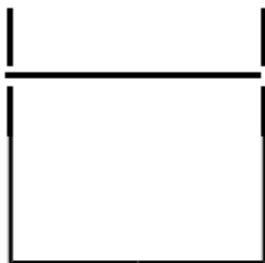


- **Temperature** is a measure of thermal energy in a substance, which is independent on the amount of matter.
 - This is an intensive property
 - Other examples of intensive properties: boiling point, melting point, color,...
- **Heat** is a form of thermal energy, which is dependent on the amount of matter.
 - This is an extensive property
 - Other examples of extensive properties: mole, mass, volume,...

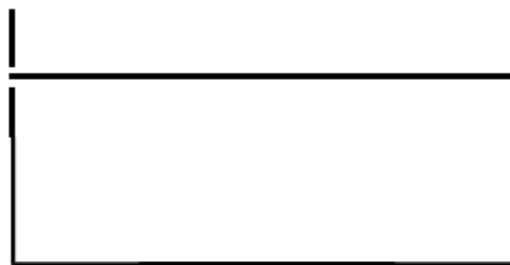
EXAMPLE

Which of the following has the greater amount of heat?

a.

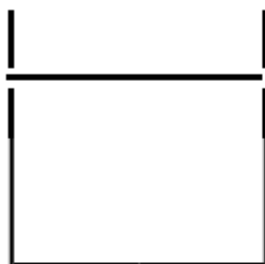


30 g H₂O at 50°C

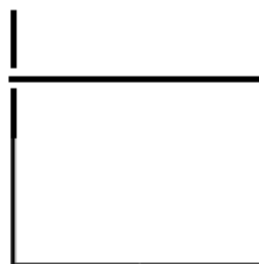


300 g H₂O at 50°C

b.

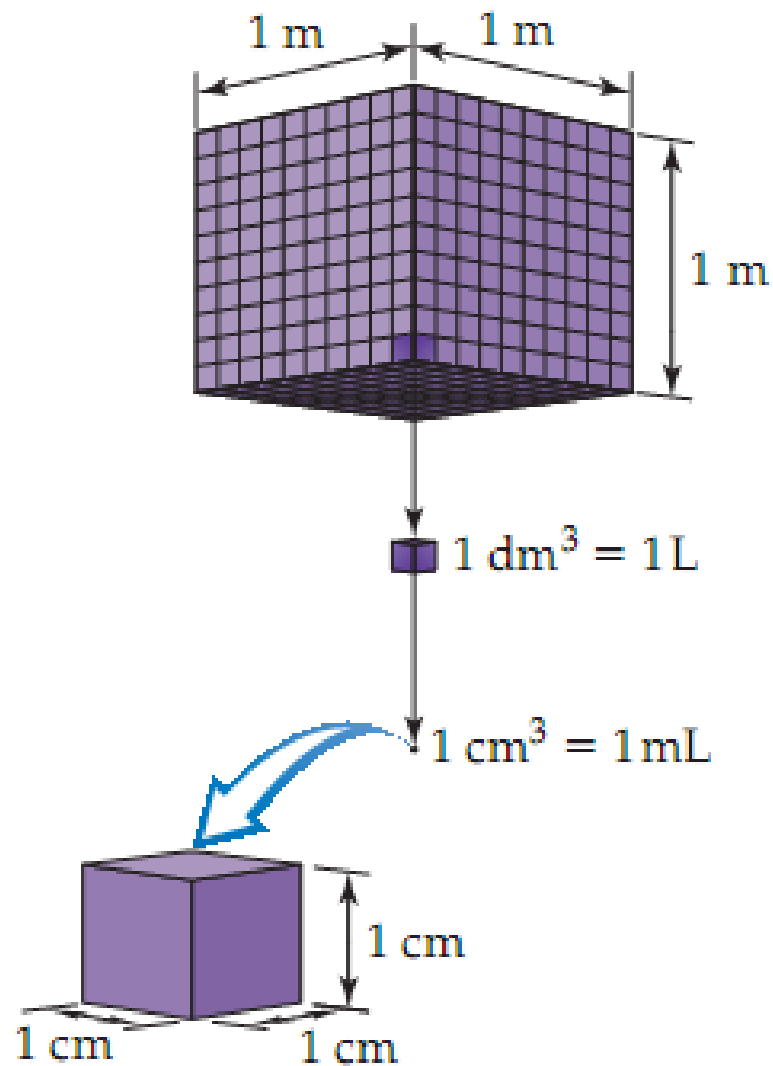


30 g H₂O at 50°C



30 g H₂O at 100°C

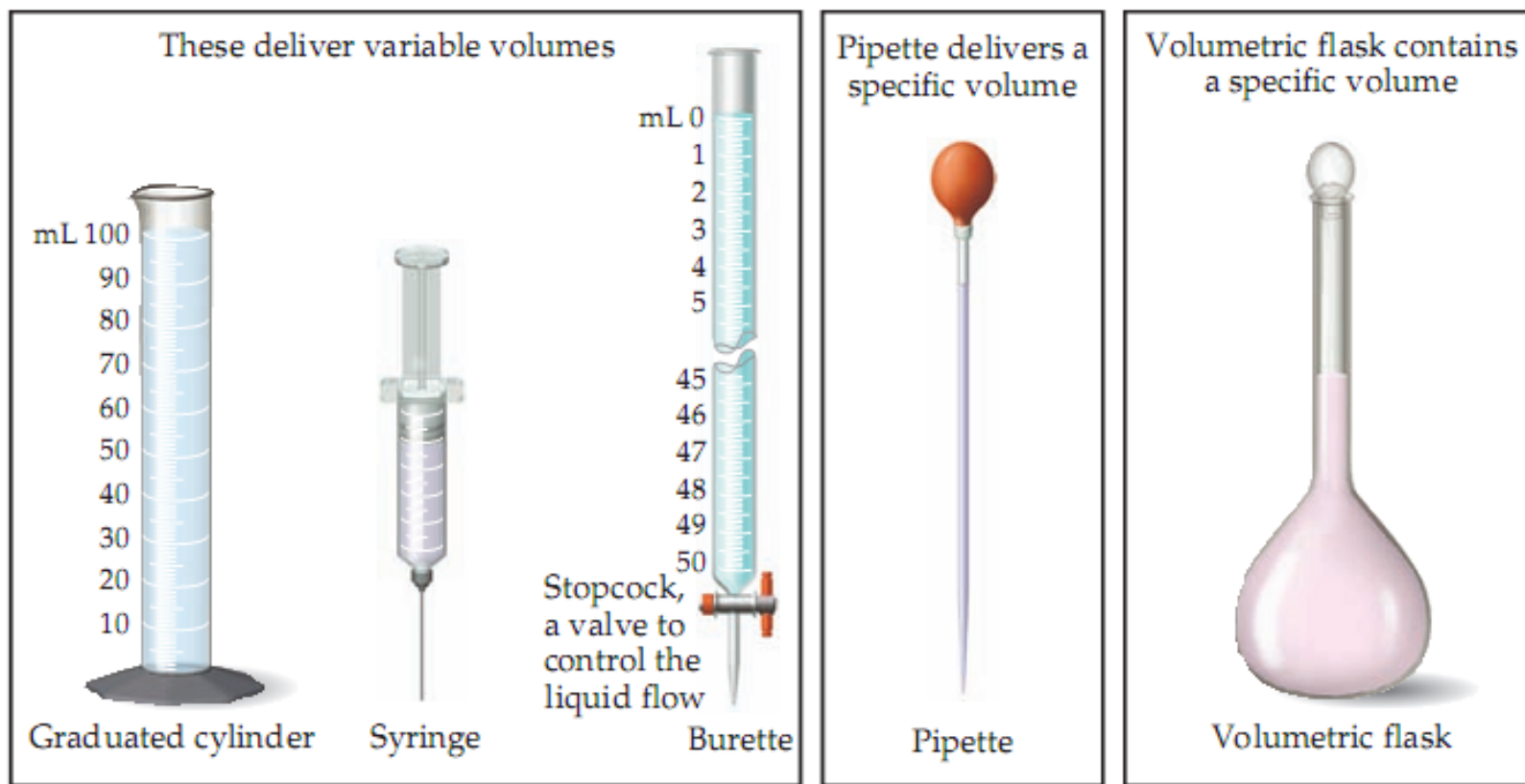
Volume



Density

Density is the amount of mass in a unit volume of a substance.

$$\text{Density} = \frac{\text{mass}}{\text{volume}}$$



Common volumetric glassware for measuring volume of liquids

SI and non-SI Units Compared

1 Imperial qt
1.136 L

1 US qt
0.936 L

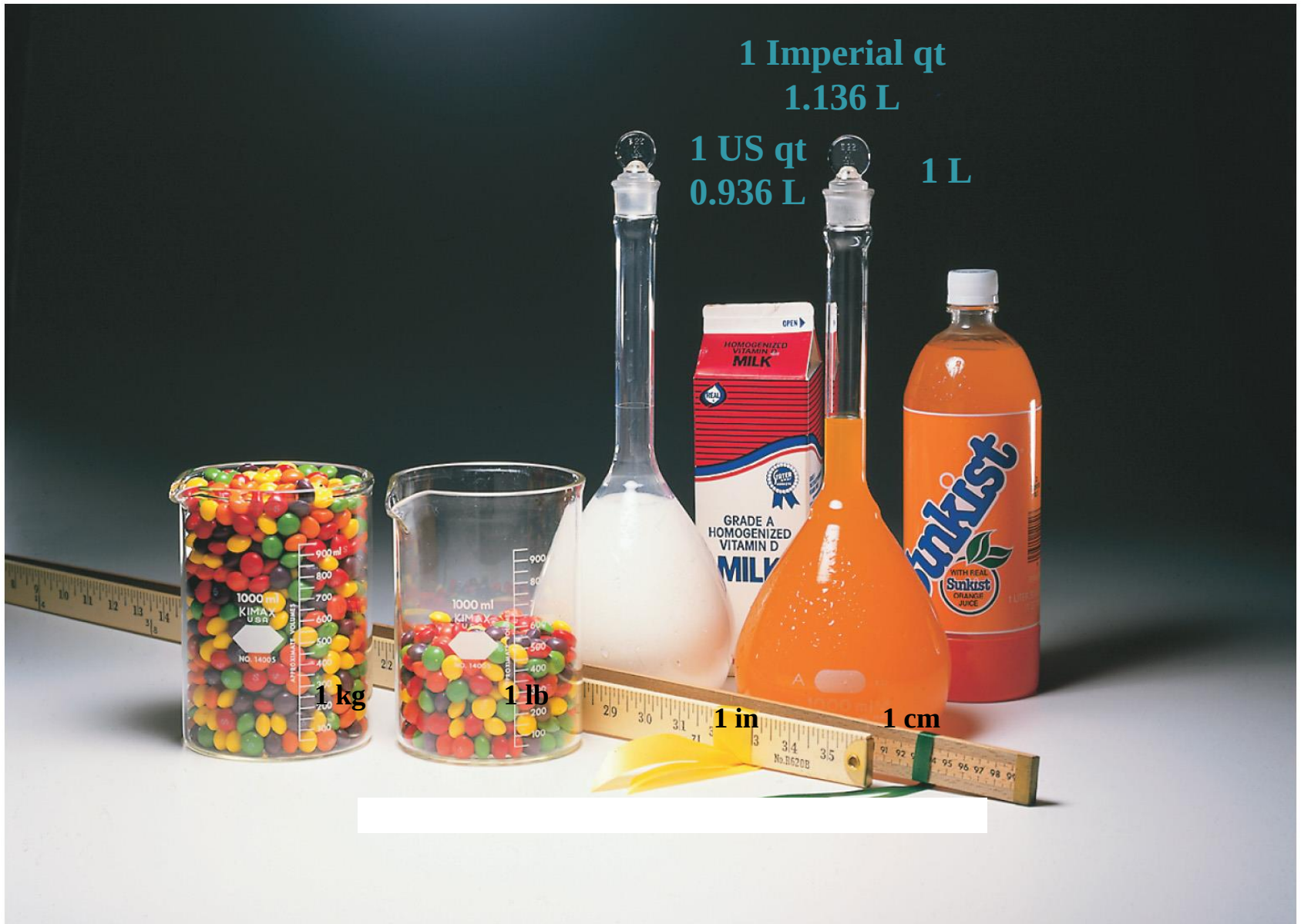
1 L

1 kg

1 lb

1 in

1 cm



Units

S.I. Units

Length	meter, m
Mass	Kilogram, kg
Time	second, s
Temperature	Kelvin, K
Quantity	Mole, $6.022 \times 10^{23} \text{ mol}^{-1}$

Other Common Units

Length	Angstrom, Å, 10^{-8} cm
Volume	Liter, L, 10^{-3} m^3
Energy	Calorie, cal, 4.184 J
Pressure	
	$1 \text{ Atm} = 1.064 \times 10^2 \text{ kPa}$
	$1 \text{ Atm} = 760 \text{ mm Hg}$

Derived Quantities

Force	Newton, kg m s^{-2}
Pressure	Pascal, $\text{kg m}^{-1} \text{ s}^{-2}$
Energy	Joule, $\text{kg m}^2 \text{ s}^{-2}$

Density and Percent Composition

$$\delta = m/V$$

$$m = V\delta$$

$$V = m/\delta$$

$$\text{g/mL}$$

Mass and volume are **extensive** properties

Density is an **intensive** property

Density in Conversion Pathways

What is the mass of a cube of osmium that is 1.25 inches on each side?

Have volume, need density = 22.48g/cm³

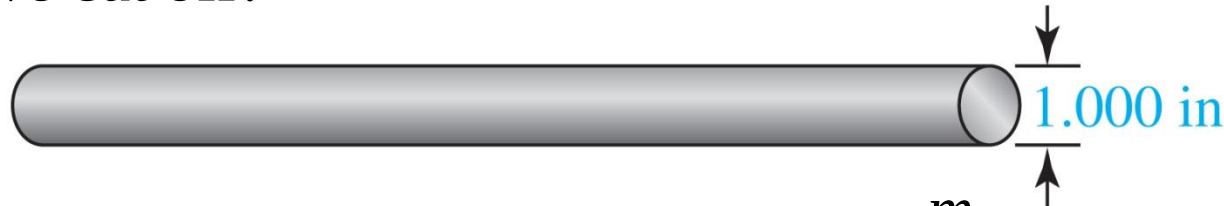
(converts in. to cm) (converts cm to cm³) (converts cm³ to g osmium)

$$? \text{ g osmium} = \left[1.25 \cancel{\text{ in.}} \times \frac{2.54 \cancel{\text{ cm}}}{1 \cancel{\text{ in.}}} \right]^3 \times \frac{22.48 \text{ g osmium}}{1 \cancel{\text{ cm}^3}} = 719 \text{ g osmium}$$

EXAMPLE

Relating Mass, Volume and Density

The stainless steel in the solid cylindrical rod pictured below has a density of 7.75 g/cm^3 . If we want a 1.00 kg mass of this rod, how long a section must we cut off?



Calculate the Volume needed

$$V = \frac{m}{d} = m \times \frac{1}{d}$$

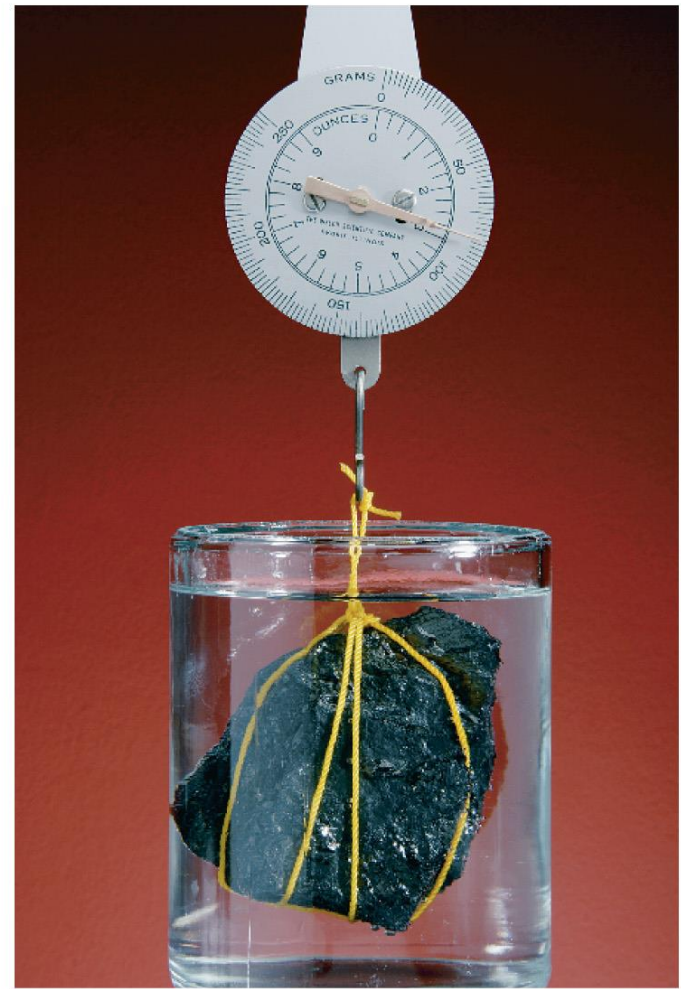
$$V = (1.00\cancel{\text{kg}})(1000\cancel{\text{g/kg}}) \times \frac{1\text{ cm}^3}{7.75\cancel{\text{g}}} = 129 \text{ cm}^3$$

Calculate the Length

$$V = h \pi r^2 \quad h = \frac{V}{\pi r^2}$$

$$h = \frac{V}{\pi r^2} = \frac{129\text{ cm}^3}{(3.1415)\{(0.5)(1.000\cancel{\text{in}})(2.54\cancel{\text{cm/in}})\}^2} = 25.5 \text{ cm}$$

Measuring Volume of an Irregular Object



1-5 Uncertainty in Measurement



Good accuracy
Good precision

- Precision
 - Reproducibility of a measurement.



Poor accuracy
Good precision

- Accuracy
 - How close to the real value.



Poor accuracy
Poor precision

- Systematic errors.
 - Constantly lower
 - Constantly higher
- Random errors
 - Limitation in reading a scale.

Precision



Reproducibility ± 0.1 g

Precision low

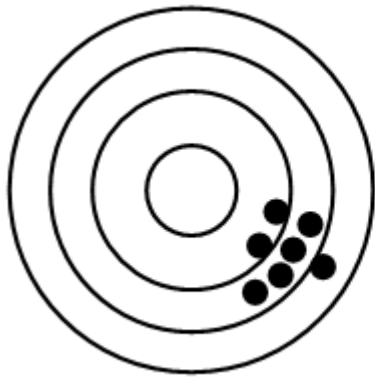


± 0.0001 g

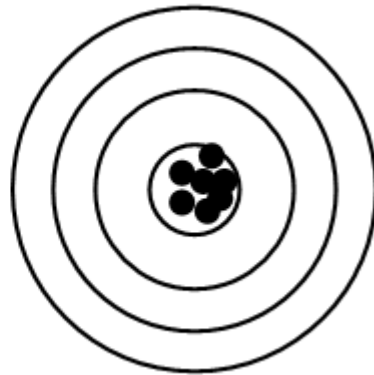
high

EXAMPLE

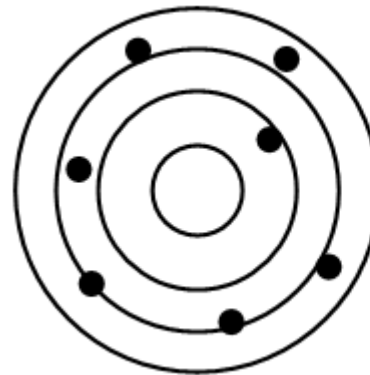
Which of the 4 following images is not precise and not accurate?



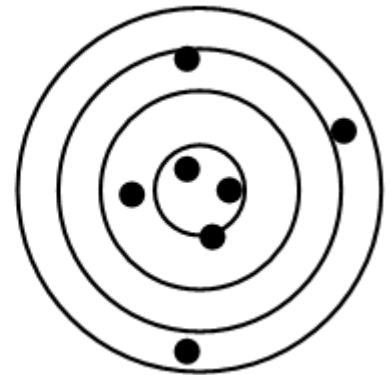
1



2



3



4

EXAMPLE

A student must measure the weight of a sodium bicarbonate compound, NaHCO_3 , and obtains the following measurements: 23.12 g, 23.08 g and 23.17g. If the true weight of the compound is 18.01 g what can be said about the student's results?

- a. *They are accurate and precise.*
- b. *They are accurate, but not precise.*
- c. *They are neither accurate or precise.*
- d. *They are not accurate, but precise.*

Standard Deviation

The standard deviation measures **how close** data results are in relation to the mean, or average, value.

$$SD = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

x_i = Observed value

$\bar{x}$ = Mean value

n = Number of observations

EXAMPLE

Calculate the standard deviation for the following results:

a. 5.29, 5.35, and 5.31

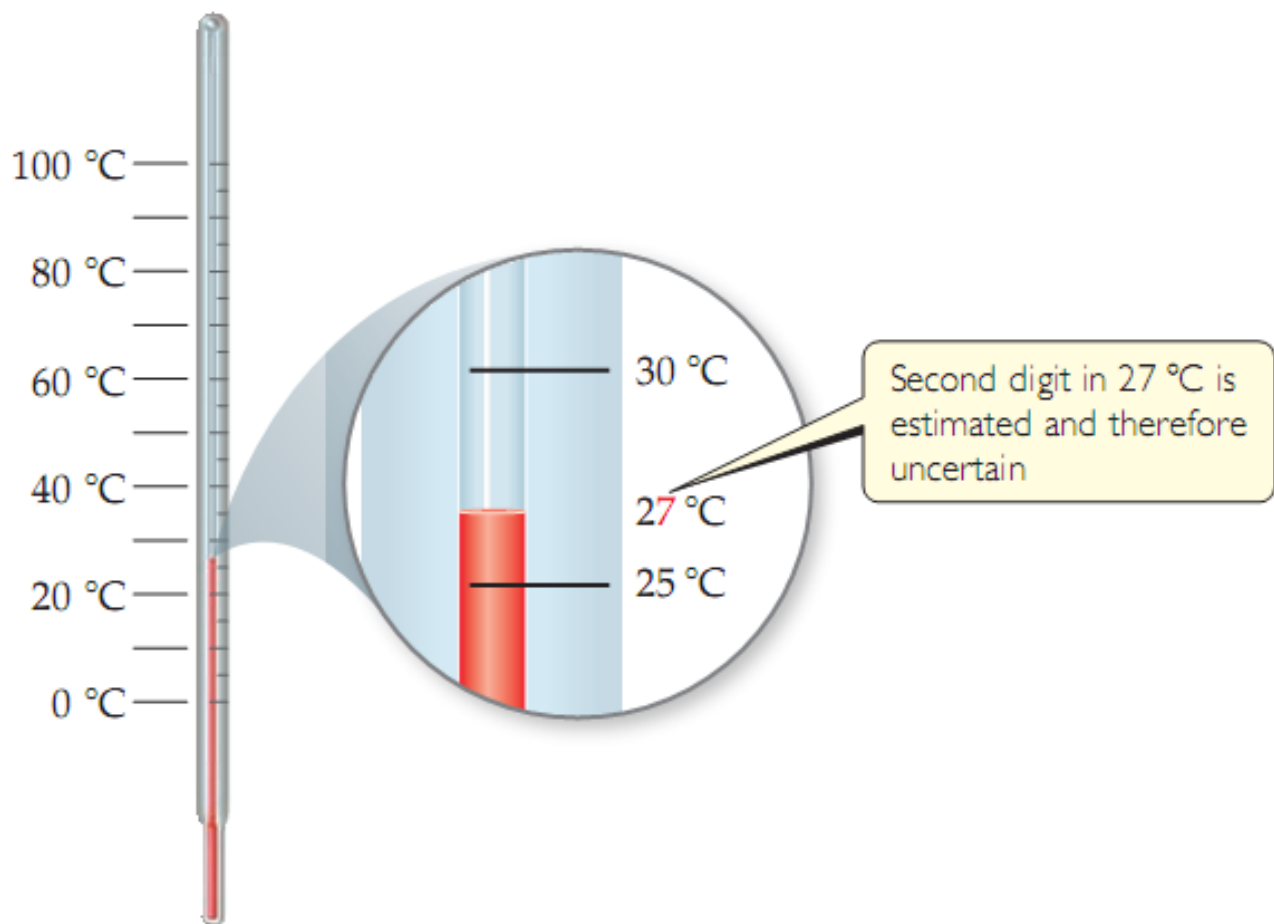
b. 0.39, 0.061, and 5.3×10^{-2}

Now, answer this question:

How tall are you?

Significant Figures

All digits of a measured quantity, including the uncertain one, are called significant figures



Significant Figures

Not significant:

zero for
“cosmetic”
purpose

0

Not significant:

zeros used only
to locate the
decimal point

0

0

4

0

0

4

5

0

0

Significant:

all nonzero
integers

Significant:

all zeros between
nonzero numbers

Significant:

zeros at the end of
a number to the right
of decimal point

EXAMPLE

How many sig figs does each number contain?

a. 0.0000185 m

b. 749 mol

c. $17.3 \times 10^3\text{ mL}$

d. 100 min

e. 0.0010050 kg

f. 1560 mol

Significant figures in calculation

For **addition and subtraction**, the result has the same number of decimal places as the measurement with the **fewest decimal places**

$$20.42 + 1.322 + 83.1 = 104.842$$

two decimal places

three decimal places

one decimal place

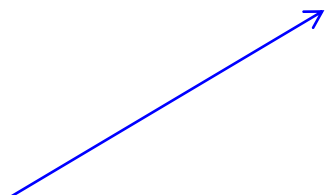
We report the result as **104.8** because 83.1 has only one decimal place

Significant figures in calculation

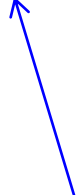
For **multiplication and division**, the result contains the same number of significant figures as the measurement with the **fewest significant figures**

$$\text{Area} = (6.221 \text{ cm})(5.2 \text{ cm}) = 32.3492 \text{ cm}^2 \Rightarrow \text{round off to } 32 \text{ cm}^2$$

four significant figures



two significant figures



We report the result as **32 cm²** because 5.2 has two significant figures

Check it at <https://www.omnicalculator.com/math/sig-fig>

EXAMPLE

Perform the following calculation to the right number of sig figs:

a. $(3.16) \times (0.003027) \times (5.7 \times 10^{-3})$

$$0.000054522324 = 5.4522324 \times 10^{-5} = 5.5 \times 10^{-5}$$

b. 2.628×10^6

$$6.281 \times 10^4$$

$$\underline{+ 0.827 \times 10^7}$$

$$10960180 = 1.0960180 \times 10^7 = 1.10 \times 10^7$$

EXAMPLE

Perform the following calculation to the right number of sig figs:

a. $(3.16) \times (0.003027) \times (5.7 \times 10^{-3})$

$= 5.5 \times 10^{-5}$

b.

2.628×10^6	0.110×10^8
6.281×10^4	1.10×10^7
<u>$+ 0.827 \times 10^7$</u>	

Significant figures in calculation

When a calculation involves two or more steps and you write answers for intermediate steps, **retain at least one nonsignificant digit** for the intermediate answers.